Notes: Edmond, Midrigan, and Xu (AER, 2015)

Competition, Markups, and the Gains from International Trade

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1 Big picture

- **Question.** Can international trade reduce product-market distortions by lowering markups and markup dispersion, and if so, how large are the resulting productivity gains?
- **Setting.** The paper studies a quantitative trade model with endogenous variable markups and calibrates it to highly disaggregated Taiwanese manufacturing data.
- **Core challenge.** In models with variable markups, trade can either *reduce* or *increase* misallocation. The effect depends on whether foreign competition disciplines dominant domestic producers or instead allows already productive firms to gain even more market share.
- **Main theoretical result.** The relevant object is not the unconditional distribution of markups but the *joint distribution of markups and employment*. Trade matters because it can shrink the markups of the firms that account for a large share of labor and sales.
- **Main empirical strategy.** The authors calibrate the model to match concentration within and across sectors, the relationship between labor shares and market shares, the aggregate import share, the fraction of exporters, and the aggregate trade elasticity.
- **Main quantitative result.** In the benchmark model, moving from autarky to Taiwan's degree of openness raises aggregate productivity by **12.4%**. Of this, **2.0%** comes from procompetitive effects, i.e., from the reduction in misallocation induced by lower markup dispersion.
- **Main mechanism.** Under autarky, the economy is about **9%** below the first best. With trade at Taiwan's observed openness, the economy is about **7%** below the first best. The main reason is that trade sharply reduces the markups of dominant producers that had been close to monopoly positions under autarky.
- **Economic takeaway.** Trade liberalization is not only a standard source of gains through specialization and variety. It can also be a powerful substitute for difficult product-market reforms because foreign competition disciplines market power.

2 Data and measurement (key objects)

2.1 Observed Taiwanese manufacturing data

- **Main dataset.** The paper uses the *Taiwan Annual Survey of Manufacturing*, covering the years **2000** and **2002–2004**. The year 2001 is missing because Taiwan ran a manufacturing census that year.
- **Two data layers.**
  - An **establishment-level** file with employment, labor expenditure, materials and energy expenditure, and total revenue.
  - A **product-level** file with revenues for each product produced by an establishment.
- **Product definition.** Products are classified at a **7-digit** Taiwanese SIC level, which is comparable to highly disaggregated U.S. product definitions.
- **Trade data.** The authors merge in detailed import data at the **HS-6** level and map them to the Taiwanese 7-digit product codes to construct import penetration rates by product.

2.2 Markets, sectors, firms, and openness

- A **sector** is a finely defined product market indexed by $s \in [0, 1]$.
- Within each sector, there are finitely many domestic and foreign producers.
- Firms compete within sectors, and sectors are then aggregated into a final good.
- The key openness moments are:
  - aggregate import share: **0.38**,
  - fraction of exporters: **0.25**,
  - aggregate trade elasticity target: **4.0**.

2.3 Key descriptive facts used in calibration

- The Taiwanese data show strong concentration both **within sectors** and **across sectors**.
- **Across sectors:**
  - the top **1%** of sectors account for about **26%** of aggregate sales and **11%** of the wage bill,
  - the top **5%** of sectors account for roughly **one-half** of sales and about **one-third** of the wage bill.
- **Across producers:**
  - the top **1%** of producers account for about **41%** of sales and **24%** of the wage bill,
  - the top **5%** of producers account for about **65%** of sales and **47%** of the wage bill.
- There is a strong negative relationship between a producer's labor share and its market share. In the data, the regression of labor share on market share yields an intercept of about $b_0 = 0.64$ and slope $b_1 = -0.50$, so the key ratio is

$$\frac{b_1}{b_0} \approx -0.78.$$

- This tells us that larger firms have systematically lower labor shares, which in the model means they have systematically higher markups.

2.4 What is observed and what is calibrated

- **Observed or targeted from the data:**
  - concentration moments,
  - import share and exporter share,
  - the regression coefficient linking labor shares to market shares,
  - the aggregate trade elasticity.
- **Calibrated parameters in the benchmark model:**

$$\gamma = 10.50, \qquad \theta = 1.24, \qquad \xi_x = 4.58, \qquad \xi_z = 0.51,$$

$$\zeta = 0.043, \qquad f_d = 0.004, \qquad f_x = 0.203, \qquad \tau = 1.129, \qquad \rho = 0.94.$$

- **Interpretation.**
  - γ is the within-sector elasticity of substitution.
  - θ is the across-sector elasticity of substitution.
  - ξ_x and ξ_z govern idiosyncratic and sectoral productivity dispersion.
  - f_d and f_x are fixed costs of domestic operation and exporting.
  - τ is the iceberg trade cost.
  - ρ captures cross-country correlation in sectoral productivity draws.

3 Models and empirical specifications (all equations + interpretation)

3.1 Final-good producers and sectoral structure

The home country produces a final good by aggregating sectoral outputs:

$$Y = \left(\int_0^1 y(s)^{\frac{\theta-1}{\theta}} ds \right)^{\frac{\theta}{\theta-1}},$$

where $\theta > 1$ is the elasticity of substitution *across sectors*.

Within each sector s, output is itself a CES aggregate of domestic and imported intermediates:

$$y(s) = \left(\sum_{i=1}^{n(s)} y_{iH}(s)^{\frac{\gamma-1}{\gamma}} + \sum_{i=1}^{n(s)} y_{iF}(s)^{\frac{\gamma-1}{\gamma}} \right)^{\frac{\gamma}{\gamma-1}},$$

where $\gamma > \theta$ is the elasticity of substitution *within a sector*.

Interpretation.

- The economy is nested: firms first compete inside sectors, and sectors then compete in final demand.
- Because $\gamma > \theta$, competition inside a sector is more intense than competition across sectors.
- This wedge between γ and θ is what makes large-market-share firms face lower demand elasticity and charge higher markups.

3.2 Intermediate producers, trade costs, and firm pricing

A home producer i in sector s has linear technology:

$$y_i(s) = a_i(s) l_i(s),$$

where firm productivity is

$$a_i(s) = z(s) x_i(s).$$

The sector component $z(s)$ is Pareto distributed across sectors, while the idiosyncratic component $x_i(s)$ is Pareto distributed within sectors.

Trade is subject to iceberg costs. For a home producer,

$$y_i(s) = y_{iH}(s) + \tau y_{iH}^*(s),$$

where $\tau \geq 1$ means that τ units must be shipped for 1 unit to arrive abroad.

Demand for domestic intermediates is

$$y_{iH}(s) = \left(\frac{p_{iH}(s)}{p(s)}\right)^{-\gamma} \left(\frac{p(s)}{P}\right)^{-\theta} Y,$$

and similarly for imported intermediates.

The aggregate and sectoral price indexes are

$$P = \left(\int_0^1 p(s)^{1-\theta} ds\right)^{\frac{1}{1-\theta}},$$

$$p(s) = \left(\sum_{i=1}^{n(s)} \phi_{iH}(s) p_{iH}(s)^{1-\gamma} + \tau^{1-\gamma} \sum_{i=1}^{n(s)} \phi_{iF}(s) p_{iF}(s)^{1-\gamma}\right)^{\frac{1}{1-\gamma}}.$$

Under Cournot competition, a firm's domestic price is a markup over marginal cost:

$$p_{iH}(s) = \frac{\varepsilon_{iH}(s)}{\varepsilon_{iH}(s) - 1} \frac{W}{a_i(s)},$$

where the demand elasticity it faces is

$$\varepsilon_{iH}(s) = \left(\omega_{iH}(s)\frac{1}{\theta} + (1 - \omega_{iH}(s))\frac{1}{\gamma}\right)^{-1},$$

and $\omega_{iH}(s)$ is the firm's market share in sectoral revenue.

Interpretation.

- A firm with a tiny market share mostly competes with other firms in its own sector, so its elasticity is close to γ and its markup is close to $\gamma/(\gamma - 1)$.
- A dominant firm competes more with the rest of the economy, so its elasticity is closer to θ and its markup is closer to $\theta/(\theta - 1)$.
- Therefore, the markup is an increasing and convex function of market share.

3.3 Operating decisions and fixed costs

A home firm operates in its domestic market only if

$$(p_{iH}(s) - W/a_i(s)) y_{iH}(s) \geq W f_d,$$

and in its export market only if

$$(p_{iH}^*(s) - W/a_i(s)) y_{iH}^*(s) \geq W f_x.$$

Interpretation.

- There are fixed costs of domestic operation and exporting.
- More productive firms are more likely to pay these fixed costs and survive, which contributes to concentration and exporter selection.

3.4 Aggregate productivity, aggregate markup, and misallocation

Aggregate output can be written as

$$Y = A\tilde{L},$$

where A is endogenous aggregate productivity and $\tilde{L}$ is labor net of fixed costs.

Aggregate productivity is a quantity-weighted harmonic mean of firm productivity:

$$A = \left(\int_0^1 \left(\sum_{i=1}^{n(s)} \frac{1}{a_i(s)} \frac{y_{iH}(s)}{Y} + \tau \sum_{i=1}^{n(s)} \frac{1}{a_i(s)} \frac{y_{iH}^*(s)}{Y} \right) ds \right)^{-1}.$$

Define the aggregate markup as

$$\mu := \frac{P}{W/A}.$$

The paper shows that μ is a **revenue-weighted harmonic mean** of firm-level markups.

Aggregate productivity can also be written as

$$A = \left(\int_0^1 \left(\frac{\mu(s)}{\mu} \right)^{-\theta} a(s)^{\theta-1} ds \right)^{\frac{1}{\theta-1}},$$

while the planner's efficient allocation yields

$$A^{efficient} = \left(\int_0^1 a(s)^{\theta-1} ds \right)^{\frac{1}{\theta-1}}.$$

Interpretation.

- If there were no markup dispersion, then $A = A^{efficient}$.
- Markup dispersion lowers productivity because relative prices are no longer aligned with relative marginal costs.
- High-productivity firms with high markups end up too small, while low-productivity firms with low markups end up too large.

3.5 Trade elasticity and incomplete pass-through

The aggregate trade elasticity is defined as

$$\sigma := \frac{d \log \left(\frac{1-\lambda}{\lambda} \right)}{d \log \tau},$$

where λ is the aggregate share of spending on domestic goods.

With variable markups, trade-cost changes are not fully passed through into relative prices because firms adjust markups. The aggregate elasticity can be written as

$$\sigma = (\gamma - \theta) \left(\int_0^1 \frac{\lambda(s)}{\lambda} \frac{1-\lambda(s)}{1-\lambda} (1+\epsilon(s))\omega(s) ds \right) + (\theta - 1) \left(\int_0^1 \frac{1-\lambda(s)}{1-\lambda} (1+\epsilon(s))\omega(s) ds \right),$$

where $\epsilon(s)$ captures the elasticity of foreign relative to home markups with respect to trade costs.

Interpretation.

- In constant-markup models, the trade elasticity is essentially the elasticity of trade flows with respect to relative prices.
- Here, incomplete pass-through lowers the trade elasticity because foreign firms gain share and raise markups while domestic firms lose share and cut markups.
- This is why the paper needs a fairly high γ to still match the empirical trade elasticity target of 4.

3.6 Calibration strategy

The calibration maps moments to parameters as follows.

- **Concentration moments** pin down ζ, ξ_x, ξ_z, and f_d.
- **Import share, exporter share, and trade elasticity** help pin down τ, f_x, and ρ.
- **The labor-share/market-share slope** disciplines the gap between θ and γ.

A key restriction is

$$\theta = \left(\frac{1}{\gamma} - \frac{b_1}{b_0} \frac{\gamma - 1}{\gamma} \right)^{-1}.$$

In the data, $b_1/b_0 = -0.78$, and the benchmark calibration chooses

$$\gamma = 10.5, \qquad \theta = 1.24.$$

Interpretation.

- The data require a large gap between γ and θ.
- That gap means large firms face much lower elasticities than small firms, and therefore charge substantially higher markups.
- The paper also checks that alternative markup estimates that purge technology heterogeneity imply very similar values of θ and γ.

3.7 Gains from trade decomposition

The paper decomposes the gain in aggregate productivity from lower trade costs into a first-best component and a procompetitive component:

$$\Delta \log A = \Delta \log A^{efficient} + \Delta \big(\log A - \log A^{efficient} \big).$$

- The first term is the standard gain from trade holding allocation efficiency fixed.
- The second term is the **procompetitive gain**: the reduction in misallocation due to lower markup dispersion.

In the benchmark calibration:

- total gain from autarky to Taiwan openness: **12.4%**,
- first-best component: **10.4%**,

- procompetitive component: **2.0%**.

Misallocation falls from about **9%** below first best under autarky to about **7%** below first best with trade. The biggest procompetitive gains occur near autarky: moving from autarky to a 10% import share yields procompetitive gains of **1.7%**, while further opening adds smaller amounts.

Interpretation.

- The main action is not in average markups per se.
- The main action is that trade cuts the markups of the dominant firms where most labor and sales are concentrated.
- This is why the unconditional markup distribution can move little, while productivity still rises meaningfully.

3.8 Why unconditional markups are a poor guide

A key result is that the unconditional distribution of markups barely changes with trade, but misallocation still falls sharply. In the benchmark model,

- under autarky, the economy is **9%** below first best,
- with trade at Taiwan openness, it is **7%** below first best,
- the ninety-ninth percentile sectoral markup falls from **5.25** under autarky to **2.22** with trade,
- the standard deviation of log sectoral markups falls from **0.31** to **0.14**.

Interpretation. Trade mainly disciplines near-monopoly sectors, not the whole distribution uniformly. Therefore, moments of the unconditional markup distribution are a poor guide to the procompetitive gains from trade.

3.9 Robustness and extensions

The paper shows that the main message survives a wide range of alternative assumptions.

Correlated idiosyncratic draws. Allowing small cross-country correlation in idiosyncratic draws ($\rho_x = 0.05$) raises procompetitive gains. Total gains remain about **12.0%**, but procompetitive gains rise to **2.4%**, and misallocation falls by almost one-third.

Capital accumulation and elastic labor supply. When the aggregate markup also distorts investment and labor supply, welfare gains are larger. With capital accumulation and Frisch elasticity 1, welfare gains are **18.1%**, of which **3.6%** are procompetitive.

Heterogeneous labor market distortions. Adding labor wedges changes the overall level of inefficiency but leaves the markup-driven procompetitive gains essentially unchanged. Total gains are about **12.2%**, with procompetitive gains still around **2%**.

Heterogeneous tariffs. Allowing sectoral tariffs with mean **0.062** and standard deviation **0.039** raises total gains to **14.6%**, while procompetitive gains remain **2.0%**.

Bertrand competition. With Bertrand instead of Cournot competition, total gains are **13.8%** and procompetitive gains are **2.5%**. Misallocation falls by about one-half rather than one-fifth.

Free entry. With free entry, total gains fall to about **6.9%** and procompetitive gains to **1.2%**. The paper stresses that this is mostly because the alternative calibration produces less initial misallocation, not because entry itself kills the mechanism.

Collusion plus free entry. If 25% of high-productivity firms collude, total gains are **11.6%** and procompetitive gains are **4.3%**. Trade is then even more valuable because imports break domestic collusion.

4 Identification and instruments (what makes the estimates credible)

4.1 What is hard to identify

- Trade gains depend on both the level of misallocation and on how responsive markups are to trade costs.
- Neither the markup distribution nor the productivity distribution is directly observed.
- Cross-country similarity in productivity is especially important: too little correlation means too little head-to-head competition and possibly even negative procompetitive effects.

4.2 How θ and γ are identified

There are no external instruments in the usual reduced-form sense. Instead, identification comes from the model's structural restrictions plus observed moments.

- The model implies a direct mapping from labor shares and market shares to markup variation.
- The ratio b_1/b_0 in the labor-share regression restricts admissible pairs (θ, γ).
- The aggregate trade elasticity target of 4 further narrows the set of plausible values, because low pass-through under variable markups requires a sufficiently high γ.

Interpretation. The same parameters that govern markup dispersion also govern incomplete pass-through and trade flows, so the calibration is tightly disciplined by both domestic and trade moments.

4.3 How concentration, fixed costs, and correlation are identified

- The distribution of producers per sector and concentration moments pin down ζ, ξ_x, and ξ_z.

- The domestic fixed cost f_d helps pin down the set of firms that operate and therefore the median producer size.
- The export fixed cost f_x helps match the exporter fraction.
- The trade cost τ helps match the aggregate import share.
- The copula parameter ρ is chosen so the model reproduces realistic trade elasticity, intraindustry trade, and the cross-sectional relation between import penetration and domestic size.

The benchmark requires a very high cross-country correlation,

$$\rho = 0.94,$$

meaning that, conditional on a sector being strong in one country, it tends also to be strong in the other. This is what delivers genuine head-to-head competition.

4.4 Why the benchmark is credible

- The model simultaneously matches domestic concentration, openness, exporter share, and the markup-size relationship.
- The paper checks that using alternative markup estimates that purge technology heterogeneity leaves the calibrated elasticities almost unchanged.
- The paper also shows that the ACR formula approximates total gains well, so the model is not producing implausible overall trade gains.

Key point. The paper's value added is not that it overturns standard total-gains formulas. It is that it shows those total gains can hide an additional and economically important *composition* effect through markups.

4.5 Main limitations

- The paper is a static trade model in the benchmark and abstracts from aggregate uncertainty.
- Markups are inferred through labor shares, so if labor shares also vary because of other distortions, the interpretation requires care.
- The benchmark assumes two symmetric countries and exogenous numbers of potential firms; these assumptions are later relaxed only in extensions.
- The data are from Taiwanese manufacturing in the early 2000s, so the quantitative magnitudes are application-specific even if the mechanism is more general.

4.6 Relation to the literature

- Relative to ACR, the paper says that total gains can still be summarized by the trade elasticity, but meaningful procompetitive gains can coexist with that fact.
- Relative to constant-markup trade models like Eaton–Kortum or Melitz, the paper makes markup dispersion central.

- Relative to misallocation papers such as Hsieh–Klenow, it endogenizes misallocation through market structure and international trade.

5 Tables and figures

5.1 Table 1: calibration moments and parameterization

Read:

- Table 1 is the core calibration table.
- The paper matches strong concentration both within and across sectors, a fraction of exporters of **0.25**, an aggregate import share of **0.38**, and a trade elasticity of **4.0**.
- The benchmark parameter values are economically revealing: $\gamma = 10.5$, $\theta = 1.24$, and $\rho = 0.94$.
- The very high ρ says the model needs strong cross-country productivity correlation to generate enough head-to-head competition while still matching the aggregate trade elasticity.

TABLE 1—PARAMETERIZATION

	Data	Model		Data	Model
Panel A. Moments					
Within-sector concentration, domestic sales			Size distribution sectors, domestic sales		
Mean inverse hh	7.25	4.30	Fraction sales by top 0.01 sectors	0.26	0.21
Median inverse hh	3.92	3.79	Fraction sales by top 0.05 sectors	0.52	0.32
Mean top share	0.45	0.46	Fraction wages (same) top 0.01 sectors	0.11	0.22
Median top share	0.40	0.41	Fraction wages (same) top 0.05 sectors	0.32	0.33
Distribution of sectoral shares, domestic sales			Size distribution producers, domestic sales		
Mean share	0.04	0.05	Fraction sales by top 0.01 producers	0.41	0.33
Median share	0.005	0.005	Fraction sales by top 0.05 producers	0.65	0.61
p75 share	0.02	0.03	Fraction wages (same) top 0.01 producers	0.24	0.31
p95 share	0.19	0.27	Fraction wages (same) top 0.05 producers	0.47	0.57
p99 share	0.59	0.59			
SD share	0.11	0.12			
Across-sector concentration			Coefficients in regression of labor share on market share		
p10 inverse HH	1.17	1.70	Ratio of coefficients b_1/b_0	−0.78	−0.78
p50 inverse HH	3.73	3.79			
p90 inverse HH	13.82	7.66	Import and export statistics		
p10 top share	0.16	0.24	Aggregate fraction exporters	0.25	0.25
p50 top share	0.41	0.41	Aggregate import share	0.38	0.38
p90 top share	0.92	0.75	Trade elasticity	4.00	4.00
p10 number producers	2	3	Coefficient, share imports on share sales	0.81	0.55
p50 number producers	10	16	Index import share dispersion	0.38	0.26
p90 number producers	52	47	Index intraindustry trade	0.37	0.45

Panel B. Parameter values		
γ	10.50	Within-sector elasticity of substitution
θ	1.24	Across-sector elasticity of substitution
ξ_x	4.58	Pareto shape parameter, idiosyncratic productivity
ξ_z	0.51	Pareto shape parameter, sector productivity
ζ	0.043	Geometric parameter, number producers per sector
f_d	0.004	Fixed cost of domestic operations
f_x	0.203	Fixed cost of export operations
τ	1.129	Gross trade cost
ρ	0.94	Kendall correlation for Gumbel copula

TABLE 2—MARKUPS IN DATA AND MODEL

		Benchmark		Bertrand	
	Data	Taiwan	Autarky	Taiwan	Autarky
Panel A. Markup moments					
Aggregate markup		1.31	1.35	1.21	1.24
Unconditional markup distribution					
Mean	1.13	1.15	1.14	1.11	1.12
p50	1.11	1.11	1.11	1.11	1.11
p75	1.12	1.13	1.12	1.11	1.11
p90	1.14	1.21	1.18	1.12	1.12
p95	1.18	1.31	1.27	1.15	1.15
p99	1.41	1.68	1.64	1.33	1.37
SD log	0.06	0.08	0.09	0.04	0.08
log p95/p50	0.06	0.17	0.14	0.04	0.04
Across-sector markup distribution					
Mean	1.29	1.37	1.54	1.22	1.26
p50	1.18	1.30	1.31	1.18	1.18
p75	1.30	1.40	1.45	1.24	1.26
p90	1.50	1.61	1.83	1.39	1.49
p95	1.77	1.84	2.50	1.54	2.17
p99	2.76	2.22	5.25	2.09	5.25
SD log	0.18	0.14	0.31	0.12	0.32
log p95/p50	0.41	0.35	0.65	0.27	0.61
Panel B. Aggregate implications					
Import share	0.38	0.38	0	0.38	0
Fraction exporters	0.25	0.25	0	0.25	0
TFP loss, percent		7.0	9.0	2.1	4.6
Gains from trade, percent		12.4	—	13.8	—
Procompetitive gains, percent		2.0	—	2.5	—

Notes: The benchmark model features Cournot competition. TFP losses are the percentage gap between the level of aggregate productivity and the first-best level of aggregate productivity associated with the planning allocation (subject to the same trade costs). The gains from trade are the percentage change in aggregate productivity relative to autarky. The procompetitive gains from trade are the percentage change in aggregate productivity less the percentage change in first-best productivity.

5.2 Table 2: markups in data and model

Read:

- Table 2 shows that the model reproduces the broad level and dispersion of markups in the Taiwanese data.
- More important, it shows why unconditional markups are misleading. Under autarky, the unconditional markup distribution barely changes, yet misallocation is much worse.
- The striking number is the sectoral ninety-ninth percentile markup: it falls from **5.25** in autarky to **2.22** with trade.

- The drop is concentrated exactly where it matters most: dominant firms and near-monopoly sectors.

5.3 Table 3: gains from trade

Read:

- Table 3 contains the headline quantitative results.
- From autarky to Taiwan openness, aggregate productivity rises by **12.4%**.
- The first-best component is **10.4%**, while the procompetitive component is **2.0%**.
- The biggest procompetitive gains come early: opening from autarky to a 10% import share already delivers **1.7%** in procompetitive gains.
- So trade liberalization does not need to be massive to generate meaningful product-market discipline.

TABLE 3—GAINS FROM TRADE

Change in import share	0 to 10	10 to 20	20 to 30	30 to Taiwan	0 to Taiwan
Panel A. Benchmark model					
Change TFP, percent	3.4	2.7	3.3	3.0	12.4
Change first-best TFP, percent	1.8	2.5	3.2	3.0	10.4
Procompetitive gains, percent	1.7	0.3	0.1	0.0	2.0
Misallocation relative to autarky	0.81	0.79	0.78	0.78	0.78
Change aggregate markup, percent	−1.9	−0.6	−0.4	−0.1	−2.9
Domestic	−1.6	−0.6	−0.4	−0.3	−2.9
Import	16.6	−0.1	0.4	0.2	17.1
Change markup dispersion, percent	−1.7	−0.2	1.1	−0.1	−0.9
Domestic	−1.9	−0.4	1.0	−0.4	−1.7
Import	10.3	−0.1	0.0	0.1	10.3
Trade elasticity (ex post)	4.2	4.1	4.0	4.0	4.0
ACR gains, percent	2.5	2.9	3.3	2.8	11.7
Panel B. Alternative model with correlated $x_i(s)$, $x_i^*(s)$					
Change TFP, percent	3.3	2.6	3.1	3.0	12.0
Change first-best TFP, percent	1.6	2.2	2.9	2.9	9.6
Procompetitive gains, percent	1.7	0.3	0.2	0.1	2.4
Misallocation relative to autarky	0.81	0.77	0.75	0.73	0.73
Change aggregate markup, percent	−2.0	−0.8	−0.5	−0.2	−3.5
Domestic	−1.7	−0.6	−0.5	−0.4	−3.2
Import	15.7	−0.2	0.3	0.3	16.1
Change markup dispersion, percent	−0.4	−1.7	0.9	−1.2	−2.4
Domestic	−0.3	−2.0	0.9	−1.7	−3.1
Import	8.1	0.0	0.1	0.2	8.4
Trade elasticity (ex post)	4.2	4.2	4.1	4.0	4.0
ACR gains, percent	2.5	2.8	3.3	2.8	11.7

Notes: Panel A shows the gains from trade for our benchmark model. Panel B shows the gains from trade for our alternative model with correlation in idiosyncratic draws $x_i(s)$, $x_i^*(s)$ chosen to match the cross-sectional relationship between import penetration and domestic producer concentration, as discussed in the main text. For our benchmark model $x_i(s)$, $x_i^*(s)$ are independent and there is cross-country correlation in productivity only through correlation in sectoral productivity $z(s)$, $z^*(s)$. Markup dispersion measured by the standard deviation of log markups.

TABLE 5—GAINS FROM TRADE WITH ELASTIC FACTORS

		Variable markups		
		Frisch elasticity of labor supply $(1/\eta)$		
	Constant markups	0	1	∞
Change TFP, percent	10.4	12.4	12.4	12.4
Change markup, percent	0	−2.9	−2.9	−2.9
Change C, percent	15.7	19.5	21.3	23.0
Change K, percent	15.7	23.0	24.8	26.6
Change Y, percent	15.7	20.1	21.9	23.7
Change L, percent	0	0	1.8	3.6
Change welfare, percent (including transition)	14.5	18.0	18.1	18.4
Procompetitive gains	0	3.5	3.6	3.9

Notes: Representative consumer has preferences $\sum_{t=0}^{\infty} \beta^t U(C_t, L_t)$ over aggregate consumption C_t and labor L_t with $U(C, L) = \log C - L^{1+\eta}/1 + \eta$. Capital is accumulated according to $K_{t+1} = (1-\delta)K_t + I_t$. Individual producers have production function $y = ak^{\alpha} l^{1-\alpha}$. We set discount factor $\beta = 0.96$, depreciation rate $\delta = 0.1$, output elasticity of capital $\alpha = 1/3$ and elasticities of labor supply η as shown.

5.4 Table 5: capital accumulation and elastic labor supply

Read:

- Table 5 adds capital accumulation and elastic labor supply.
- Once the aggregate markup also distorts labor and investment, welfare gains become larger than the benchmark productivity gains.
- With Frisch elasticity 1, welfare gains are **18.1%**, of which **3.6%** are procompetitive.
- This is important because it shows that the benchmark's 12.4% productivity gain is not the full welfare story when factors are elastic.

5.5 Table 6: robustness

Read:

- Table 6 shows the mechanism is robust.
- Heterogeneous labor distortions, heterogeneous tariffs, alternative values of γ, and alternative copulas do not overturn the result that trade reduces misallocation.
- The magnitudes move, but the sign and economic relevance of the procompetitive effect remain.

TABLE 6—ROBUSTNESS EXPERIMENTS

	Benchmark	Labor wedge	Tariffs	Bertrand	Low γ	High γ	No fixed	Gaussian
Trade elasticity	4.00	4.00	4.00	4.00	3.59	4.00	4.00	4.00
Import share	0.38	0.38	0.38	0.38	0.38	0.38	0.38	0.38
Fraction exporters	0.25	0.25	0.25	0.25	0.25	0.25	1	0.25
TFP loss autarky	9.0	8.8	9.0	4.6	4.9	11.3	8.9	9.8
TFP loss Taiwan	7.0	6.8	6.9	2.1	2.3	9.9	6.9	7.2
Gains from trade	12.4	12.2	14.6	13.8	16.6	11.8	11.8	11.6
Procompetitive gains	2.0	2.0	2.0	2.5	2.7	1.4	2.0	2.6
Key parameters								
ρ	0.94	0.93	0.94	0.91	1.00	0.92	0.95	0.99
τ	1.129	1.129	1.055	1.132	1.080	1.138	1.136	1.129
f_x	0.203	0.065	0.055	0.110	1.350	0.040	0	0.070
Additional moments								
Avg./agg. labor share	1.16	1.35	1.17	1.09	1.07	1.23	1.19	1.20
Mean tariff			0.062					
SD tariff			0.039					

TABLE 7—GAINS FROM TRADE WITH ASYMMETRIC COUNTRIES

	Benchmark		Panel A. Larger trading partner			
			$L^* = 2L$		$L^* = 10L$	
	Home	Foreign	Home	Foreign	Home	Foreign
ρ	0.94	0.94	0.94	0.94	0.96	0.96
Trade elasticity	4.00	4.00	4.00	4.19	4.00	4.30
Import share	0.38	0.38	0.38	0.21	0.38	0.05
Fraction exporters	0.25	0.25	0.25	0.11	0.25	0.05
TFP loss autarky	9.0	9.0	9.0	9.0	9.0	9.0
TFP loss	7.0	7.0	7.3	6.8	7.1	7.5
Gains from trade	12.4	12.4	12.0	6.5	12.0	2.2
Procompetitive gains	2.0	2.0	1.7	2.1	1.9	1.4
			Panel B. More productive trading partner			
			$\bar{A}^* = 2\bar{A}$		$\bar{A}^* = 10\bar{A}$	
			Home	Foreign	Home	Foreign
ρ			0.86	0.86	0.60	0.60
Trade elasticity			4.00	3.12	4.00	1.30
Import share			0.38	0.21	0.38	0.07
Fraction exporters			0.25	0.10	0.25	0.03
TFP loss autarky			9.0	9.0	9.0	9.0
TFP loss			7.4	7.0	7.5	8.6
Gains from trade			15.3	8.1	31.9	5.6
Procompetitive gains			1.5	1.8	1.5	0.4

5.6 Table 7: asymmetric countries

Read:

- Table 7 shows that when trading partners differ in size or productivity, total gains and procompetitive gains need not move together.
- If the foreign country is much larger, the smaller home country still enjoys substantial total and procompetitive gains, while the large country gains less from integrating with a small partner.
- If the foreign country is much more productive, total gains become very large, but procompetitive gains become smaller because there is less head-to-head competition.
- This is a subtle and useful lesson: trade can deliver huge standard gains without delivering equally huge product-market-discipline gains.

5.7 Table 8: free entry and collusion

Read:

- Table 8 shows that free entry by itself weakens the benchmark procompetitive effect because the alternative calibration generates less initial sectoral markup dispersion.
- But once domestic collusion is added, procompetitive gains become even larger than in the benchmark model.
- With 25% collusion, total gains are **11.6%** and procompetitive gains are **4.3%**.

TABLE 8—ENTRY AND COLLUSION

	Panel A. No collusion			Panel B. 25 percent collusion		
	No entry	Free entry	Autarky	No entry	Free entry	Autarky
Number of firms trying to enter	187	168	187	162	160	162
TFP loss	2.0	2.2	3.4	4.6	4.6	9.0
Total fixed costs	0.45	0.41	0.45	0.31	0.30	0.31
Aggregate profits	0.45	0.46	0.51	0.38	0.38	0.41
Aggregate markup	1.25	1.26	1.32	1.27	1.27	1.35
Expected profits, entrants	0.22	0.24	0.24	0.19	0.19	0.19
Gains from trade	8.2	6.9		11.7	11.6	
Procompetitive gains	1.5	1.2		4.4	4.3	
Misallocation relative to autarky	0.57	0.64		0.51	0.52	
Change markup, percent	−5.8	−5.0		−6.3	−6.3	

- This is intuitive: if domestic firms collude, foreign competition becomes a particularly strong source of market discipline.

6 Short summary

- **Method.** The paper calibrates a variable-markup trade model with oligopolistic competition to Taiwanese manufacturing data.
- **Identification.** The key empirical discipline comes from concentration moments, the relationship between labor shares and market shares, the exporter share, the import share, and the aggregate trade elasticity.
- **Application.** The benchmark calibration implies strong within- and across-sector concentration, highly dispersed sectoral markups, and substantial initial misallocation under autarky.
- **Main result.** Trade raises aggregate productivity by **12.4%** relative to autarky and reduces misallocation by about one-fifth, generating procompetitive gains of about **2.0%**.
- **Economic message.** Trade can function as a product-market reform because it disciplines dominant firms and compresses markup dispersion where employment is concentrated.

7 Conclusion

7.1 Core conclusion

- The paper shows that international trade can substantially reduce markup distortions.
- These procompetitive gains are quantitatively meaningful and distinct from the standard gains from trade.
- The effect comes from reallocating market share away from dominant high-markup firms toward a more competitive industry structure.

7.2 Economic intuition

- Under autarky, dominant producers can become near monopolists in some sectors and charge very high markups.

- Opening to trade exposes exactly these firms to foreign competitors, cutting the markups of firms that matter most for aggregate productivity.
- What matters is not whether the *average* markup moves a lot, but whether the high-markup firms that employ many workers lose market power.

7.3 Takeaway

This paper is important because it bridges the trade and misallocation literatures. It shows that trade policy is not only about comparative advantage and variety; it is also about product-market discipline. In the Taiwanese application, that discipline is strong enough to generate sizeable productivity gains even when standard total-gains formulas continue to work well.